

TrES-3b

R-0584

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_230715 · created 2026-07-24T19:02:15+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460140.8114 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.191 +/- 0.057
Transit depth (Rp/Rs)^2	3.67 +/- 2.2 [%]
Orbital Inclination (inc)	82.49 +/- 2.87
Airmass coefficient 1 (a1)	1.37 +/- 0.49
Airmass coefficient 2 (a2)	-0.27 +/- 0.78
Scatter in the residuals of the lightcurve fit is	35.25 %
Best Comparison Star	#2 - [538, 304]
Optimal Method	PSF photometry
Transit Duration (day)	0.023 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.191 ± 0.057 vs 0.16309 (deviation 0.49σ)
Duration fit vs published	0.023 d vs 0.0567 d (deviation 2.81σ)
Mid-time O–C	-5.2 min
Depth SNR	3.4
Residual scatter	35.25 %

Light curve

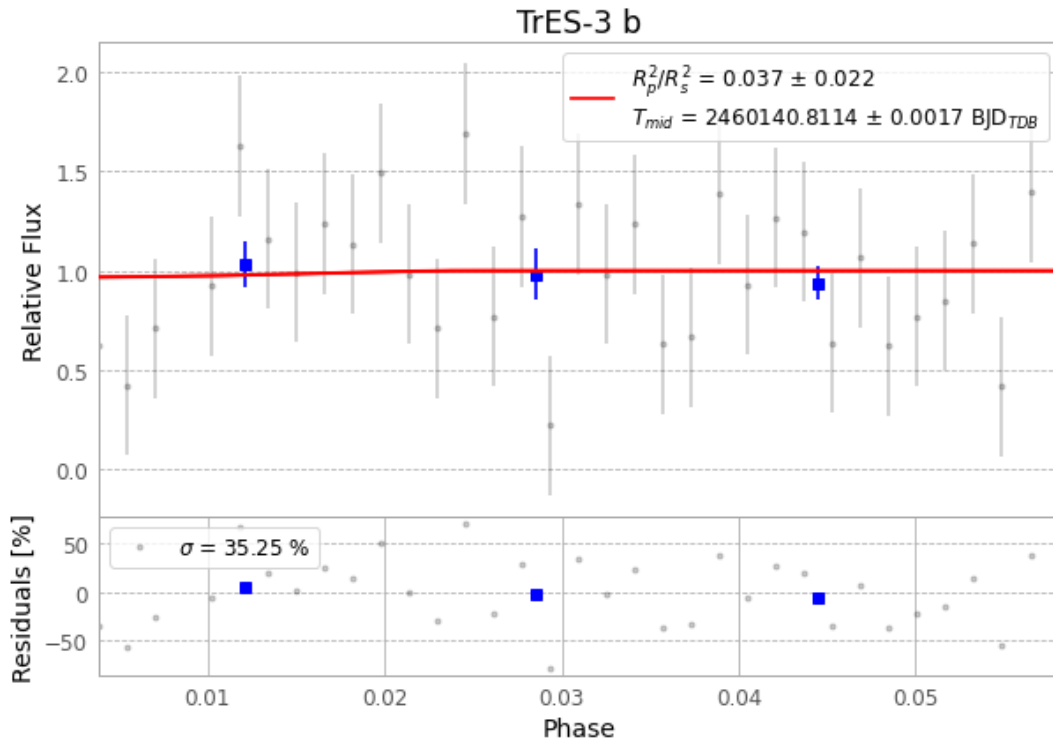


Figure 1 — FinalLightCurve_TrES-3 b_2023-07-15.png (SHA-256 b700b11cfb1e9a06...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [362, 165], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 832ac8ab5df542e3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

35 FITS frames (23 MB), TRES-3230715073018.FITS → TRES-3230715091215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-230715.log (a71b42864f9d...), FinalLightCurve_TrES-3 b_2023-07-15.png (b700b11cfb1e...), FinalLightCurve_TrES-3 b_2023-07-15.csv (9b01bc28521b...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-24 19:02Z.